HSC Mathematics Extension 1 & 2: Functions Practice

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A function is a rule that assigns to each element of one set exactly one element of another set.

—Mathematical Folklore

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1 Introduction

1.1 Project Overview

This booklet is a growing collection of functions problems for NSW HSC Mathematics Extension 1 and Extension 2 students. It covers the major themes of the syllabus: understanding what a function is, working with inverses, composing functions, and analysing their graphs. The problems are designed to build fluency with algebraic manipulation and geometric reasoning side by side.

1.2 Target Audience

This resource is written for students who want extra practice with functions in a clear, classroom-friendly format. Solutions are intentionally concise, but each one highlights the key idea—whether that is a domain restriction, a symmetry argument, or a composition rule. Both Extension 1 and Extension 2 students will find material at an appropriate level.

1.3 How to Use This Booklet

- Read the short summary of the core definitions and ideas first.
- Attempt each problem before reading the solution.
- Pay attention to domains and ranges: they are often where mistakes happen.
- Look for symmetry: the graph of an inverse is the reflection of the original in the line $y = x$.
- Use the takeaways to build pattern recognition for future HSC questions.

1.4 Extension 1 Knowledge Needed

Most problems in this booklet are accessible with standard HSC Mathematics Extension 1 knowledge. The key background ideas are:

- **Function notation:** writing $f(x)$, understanding that f is the rule and $f(x)$ is the value, and reading expressions such as $f(a + h)$ or $f(f(x))$.
- **Domain and range:** identifying the largest possible domain (the natural domain) and the corresponding range, and applying domain restrictions when stated.
- **One-to-one functions:** recognising that a function has an inverse if and only if it passes the horizontal line test.
- **Inverse functions:** swapping x and y to find f^{-1}, and understanding that $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.
- **Graphs of inverse functions:** reflecting the graph of f in the line $y = x$ to obtain f^{-1}.
- **Composite functions:** forming $f(g(x))$ by substituting one function into another, and identifying the correct domain of the composition.
- **Even and odd functions:** f is even if $f(-x) = f(x)$ for all x (the graph has y-axis symmetry); f is odd if $f(-x) = -f(x)$ for all x (the graph has rotational symmetry about the origin).
- **Transformations:** translations, reflections, dilations, and their effect on the equation and graph of a function.

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1.5 Core Formulas and Ideas

Definition of a function. A function $f : A \to B$ assigns to every element $x \in A$ exactly one element $f(x) \in B$. The set A is called the **domain** and the set B the **codomain**. The **range** is the set of all actual output values:

$$\text{range}(f) = \{f(x) : x \in A\} \subseteq B.$$

Vertical line test. A curve in the plane is the graph of a function if and only if every vertical line meets the curve at most once.

Horizontal line test. A function is one-to-one (injective) if and only if every horizontal line meets its graph at most once. Only one-to-one functions have an inverse that is also a function.

Inverse function. If f is one-to-one with domain A and range B, then its inverse $f^{-1} : B \to A$ satisfies

$$f^{-1}(f(x)) = x \quad \text{for all } x \in A, \qquad f(f^{-1}(y)) = y \quad \text{for all } y \in B.$$

To find f^{-1}: write $y = f(x)$, swap x and y, then solve for y.

Reflection principle. The graph of f^{-1} is the reflection of the graph of f in the line $y = x$.

Composite function. Given $f : B \to C$ and $g : A \to B$, the composite $f \circ g : A \to C$ is defined by

$$(f \circ g)(x) = f(g(x)).$$

Note that in general $f \circ g \neq g \circ f$.

Domain of a composite. For $f(g(x))$ to be defined, x must be in the domain of g *and* $g(x)$ must be in the domain of f:

$$\text{dom}(f \circ g) = \{x \in \text{dom}(g) : g(x) \in \text{dom}(f)\}.$$

Even and odd functions.

$$f \text{ is even} \iff f(-x) = f(x) \text{ for all } x. \qquad f \text{ is odd} \iff f(-x) = -f(x) \text{ for all } x.$$

1.6 Key Transformations

Given a base function $y = f(x)$, the standard transformations are:

- $y = f(x) + c$: shift up by c (down if $c < 0$).
- $y = f(x - h)$: shift right by h (left if $h < 0$).
- $y = af(x)$: vertical dilation by factor $|a|$; reflection in x-axis if $a < 0$.
- $y = f(bx)$: horizontal dilation by factor $\frac{1}{|b|}$; reflection in y-axis if $b < 0$.

These transformations combine: $y = af(bx - h) + c$ is the general form.

1.7 Piecewise-Defined Functions

A piecewise function assigns a different rule to different parts of the domain. For example:

$$f(x) = \begin{cases} x^2 & x < 0, \\ 2x + 1 & x \geq 0. \end{cases}$$

Key skills are: evaluating at a given x, finding the range piece by piece, and drawing the graph with correct open/closed endpoints.

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1.8 Absolute Value as a Piecewise Function

$$|x| = \begin{cases} x & x \geq 0, \\ -x & x < 0. \end{cases}$$

The graph of $y = |f(x)|$ reflects any portion of $y = f(x)$ that lies *below* the x-axis up above it. The graph of $y = f(|x|)$ keeps the right half of $y = f(x)$ and reflects it to create an even function.

1.9 Continuity and Limits (Extension 2 Note)

For Extension 2 students, a function is **continuous** at $x = a$ if

$$\lim_{x \to a} f(x) = f(a).$$

Piecewise functions need this checked at the break-points. Many function questions at Extension 2 level involve verifying continuity or using limits to determine parameters.

1.10 Problem-Solving Habits for This Booklet

- Always state the domain and range of your answer, not just the rule.
- When finding an inverse, restrict the domain of f first if the function is not one-to-one.
- When composing, work from the inside out: compute $g(x)$ first, then apply f.
- Sketch the graph whenever the question involves domains, ranges, or inverses.
- In even/odd questions, substitute $-x$ and simplify fully before drawing a conclusion.

Remark 1.1

The most common functions mistake is forgetting to adjust the domain when restricting a function to make it one-to-one before finding its inverse. Always check that your inverse has the correct domain and range.

Remark 1.2 (Notation warning)

$f^{-1}(x)$ means the inverse *function*, not $\dfrac{1}{f(x)}$. If you mean the reciprocal, write $[f(x)]^{-1}$ or $\dfrac{1}{f(x)}$ explicitly.

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2 Part 1: Worked Problems

2.1 Basic

2.2 Medium

Problem 2.1: The Nested Quadratic Chain

Consider the quadratic function

$$f(x) = x^2 - 6x + 7.$$

Find all real values of x satisfying

$$f(f(x)) = -1.$$

Hint: Do not expand $f(f(x))$ into a quartic straight away.

- Let $u = f(x)$ and solve $f(u) = -1$ first.
- Then set $f(x) = u$ for each value of u you find.
- Completing the square is a clean way to solve the resulting quadratics.

Solution 2.1

Let $u = f(x)$. Then $f(f(x)) = -1$ becomes

$$f(u) = -1,$$

so

$$u^2 - 6u + 7 = -1 \quad \Longrightarrow \quad u^2 - 6u + 8 = 0 \quad \Longrightarrow \quad (u-4)(u-2) = 0.$$

Hence

$$u = 4 \quad \text{or} \quad u = 2.$$

So we solve the two equations:

Case 1: $f(x) = 4$

$$x^2 - 6x + 7 = 4 \quad \Longrightarrow \quad x^2 - 6x + 3 = 0.$$

Thus

$$x = \frac{6 \pm \sqrt{36-12}}{2} = 3 \pm \sqrt{6}.$$

Case 2: $f(x) = 2$

$$x^2 - 6x + 7 = 2 \quad \Longrightarrow \quad x^2 - 6x + 5 = 0 \quad \Longrightarrow \quad (x-1)(x-5) = 0.$$

So

$$x = 1 \quad \text{or} \quad x = 5.$$

Therefore, the real solutions are

$$\boxed{x = 1,\ 5,\ 3 - \sqrt{6},\ 3 + \sqrt{6}.}$$

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Takeaways 2.1

- **Level-by-level solving:** Although $f(f(x))$ would expand to a quartic, it is much better to treat it as a quadratic in disguise by introducing a substitution.
- **Composition structure:** For a composite equation $f(g(x)) = c$, solve the outer layer first, then feed those answers back into the inner layer.
- **Root count intuition:** A quadratic composed with another quadratic can produce up to four real solutions, which is exactly what happens here.
- **Higher-ed connection:** Problems involving $f(f(x))$, $f(f(f(x)))$, and beyond are the first step toward iterated functions, dynamical systems, and fixed-point ideas.

2.3 Advanced

Problem 2.2: The Exponential Functional Equation

Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x) > 0$ for all real numbers x and

$$f(x+y) = f(x)f(y) \qquad \text{for all } x, y \in \mathbb{R}.$$

(i) Show that $f(0) = 1$ and deduce that $f(-x) = \dfrac{1}{f(x)}$.

(ii) By using mathematical induction, show that $f(nx) = [f(x)]^n$ for all $n \in \mathbb{Z}^+$.

(iii) Prove that $f(rx) = [f(x)]^r$ for all rational numbers r.

(iv) Suppose that f is continuous on $\mathbb{R}$. Prove that $f(tx) = [f(x)]^t$ for all real numbers t. Hence deduce that $f(x) = a^x$ for some constant a.

Hint: This is a variation of Cauchy's functional equation and is the standard route to exponential functions.

- For (i), substitute $y = 0$, then choose $y = -x$.
- For (ii), write $f((k+1)x)$ as $f(kx + x)$.
- For (iii), let $r = \dfrac{p}{q}$ and compare $f(px)$ with $f\left(q \cdot \dfrac{p}{q} x\right)$.
- For (iv), approximate a real number t by rationals $r_n \to t$ and use continuity to pass to the limit.

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Solution 2.2

(i) From $f(x) = f(x+0) = f(x)f(0)$ and $f(x) > 0$, we get $f(0) = 1$. Then

$$1 = f(0) = f(x + (-x)) = f(x)f(-x),$$

so $f(-x) = \dfrac{1}{f(x)}$.

(ii) Induct on $n \in \mathbb{Z}^+$. The case $n = 1$ is trivial. If $f(kx) = [f(x)]^k$, then

$$f((k+1)x) = f(kx + x) = f(kx)f(x) = [f(x)]^{k+1}.$$

Hence $f(nx) = [f(x)]^n$ for all $n \in \mathbb{Z}^+$.

(iii) Using (i) and (ii), the formula extends to all integers:

$$f(0x) = f(0) = 1 = [f(x)]^0, \qquad f(-nx) = \frac{1}{f(nx)} = [f(x)]^{-n}.$$

Now let $r = \dfrac{p}{q}$ with $p \in \mathbb{Z}$ and $q \in \mathbb{Z}^+$. Then

$$[f(x)]^p = f(px) = f\left(q \cdot \frac{p}{q}x\right) = \left[f\left(\frac{p}{q}x\right)\right]^q.$$

Since $f > 0$, taking the positive qth root gives

$$f\left(\frac{p}{q}x\right) = [f(x)]^{p/q}.$$

Thus $f(rx) = [f(x)]^r$ for all $r \in \mathbb{Q}$.

(iv) Let $t \in \mathbb{R}$ and choose rationals $r_n \to t$. By continuity of f,

$$f(tx) = \lim_{n\to\infty} f(r_n x).$$

Using (iii),

$$f(r_n x) = [f(x)]^{r_n}.$$

Since $f(x) > 0$, the map $u \mapsto [f(x)]^u$ is continuous, so

$$f(tx) = \lim_{n\to\infty} [f(x)]^{r_n} = [f(x)]^t.$$

Now put $x = 1$. If we let $a = f(1) > 0$, then

$$f(t) = f(t \cdot 1) = [f(1)]^t = a^t.$$

Hence $f(x) = a^x$ for some constant $a > 0$.

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Takeaways 2.2

- **Foundation of exponentials:** Once continuity is added, the functional equation $f(a+b) = f(a)f(b)$ forces the function to be exponential. In particular, if $f(1) = c$, then part (iv) shows that $f(t) = c^t$ for every real number t.
- **Variation of Cauchy's functional equation:** This is the multiplicative analogue of the classical additive equation. A common linear version says that if a continuous function $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x) + f(y) = f(x+y)$ for all real numbers x and y, then $f(x) = ax$ for some constant a.
- **Density plus continuity:** The key step is to prove the rule first for integers, then rationals, and finally extend it from $\mathbb{Q}$ to all real numbers by using the density of the rationals together with continuity.
- **Difficulty note:** This is an olympiad-style problem. You may find a variation of it in a strong trial exam from a top school, but it is unlikely to appear in an official HSC Mathematics Extension 1 or Extension 2 examination in this full form.

3 Part 2: Practice and Challenge Bank

3.1 Warm-Up Drills

3.2 Stretch Problems

3.3 Challenge Corner

4 Conclusion

Functions are one of the most unifying ideas in all of mathematics. Mastering the language of functions—domain, range, composition, and inverse—gives you the tools to read and write mathematics fluently across every topic in HSC Extension 1 and Extension 2. As this booklet grows, it will become a compact reference for the patterns and techniques that appear most often in HSC examinations.

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Repository: https://github.com/vuhung16au/math-olympiad-ml/tree/main/HSC-Functions